2023 IEEE Conference on Artificial Intelligence (CAI) | 979-8-3503-3984-0/23/$31.00 ©2023 IEEE | DOI: 10.1109/CAI54212.2023.00082

Zio: an artificial intelligence digital twin to build virtual markets

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***Abstract*—Understanding consumers' needs and how to approach them is a challenge in the digital era. The opportunity to test the best marketing strategies before launching them to the real market is a decisive advantage for a brand. To this end, we created Zio, a digital twin based on artificial intelligence. Zio is an online app that replicates, validates, simulates, and optimizes policies in competitive markets. Zio allows the management of virtual markets by means of an agent-based simulation model that represents virtual consumers with their positioning with respect to the existing brands. Evolutionary algorithms are used to auto-calibrate the model's parameters before running what-if scenarios as well as to optimize marketing strategies.**

***Index Terms*—agent-based modeling, marketing, consumer behavior heuristics, evolutionary algorithms, network science.**

I. Introduction

THE use of artificial intelligence (AI) on marketing enables more efficient decision-making process and provides marketing managers with information and insights that they could not develop in any other way [1]. There has been significant academic research into examples of how AI can facilitate marketing problems. A good set among them is focused on the relationships among consumers and how their behavior emerges to form a market [2]. In a nutshell, the goal is to recreate a real market in a model as a digital twin to understand the market dynamics and successfully apply policies before running them into the real market [3].

One of the most convenient AI techniques to represent a market of consumers is agent-based modeling (ABM) [4]. ABM is a bottom-up simulation technique where agents have heterogeneous properties and actions, interact with their peers and environment, and are able to model emergent phenomena [2]. ABM is the core technique of our AI digital twin for virtual markets, called Zio (www.zio-analytics.com). It is an operating Web application where multiple users can operate online and collaboratively. Zio allows us to: a) create virtual markets with specific parameterization, b) auto-calibrate the internal dynamics of the market with respect to historical data to validate the model, c) launch what-if scenarios from a validated market, and d) run optimization processed to assist finding out the best possible marketing strategies.

Research funded by MCIN/AEI/10.13039/501100011033/, Andalusian Gov. and FEDER Una manera de hacer Europa under grants CONFIA (PID2021-122916NB-I00) and Emergia (EMERGIA21-00139).

Apart from relying on agent-based simulations, the AI-based simulation engine in Zio includes artificial social networks to describe the information diffusion processes and word-of-mouth phenomena [5]. Additionally, Zio integrates a calibration algorithm [6], also called parameter estimation process, where an evolutionary algorithm (EA) [7] is used to search for the best possible set of parameters' values to adjust the behavior and output of the model to historical data.

Figure 1 shows the main AI technologies used. Zio permits managing all the created digital twins of real competitive markets with a database system and a user interface. Managers can save markets and reports from the simulation outputs, generate what-if scenarios, and launch required optimization processes. Zio has been validated computationally and by many clients from different sectors as Coca Cola, KFC, and Domino's Pizza (food), WiZink (bank), Jaguar-Land Rover (auto), L'Oréal and LVMH (luxury), Havas Media (media), Samsung (tech), Iberia (airlines), Naturgy (energy), and Telefonica/O2 (telco).

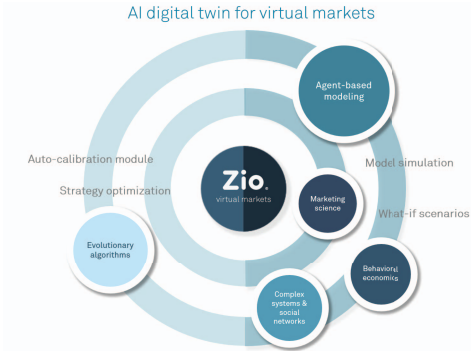

Fig. 1. General scheme of Zio digital twin and its AI technologies.

II. Main modules of Zio

A. A market of virtual consumers

A population of virtual consumers Z in a competitive market of B brands is generated to replicate the real consumers. The properties or parameters for each agent i are set up heterogeneously from surveys, marketing track data, and focus groups. There are three main properties of consumers: i) each agent has a brand awareness [8] for each

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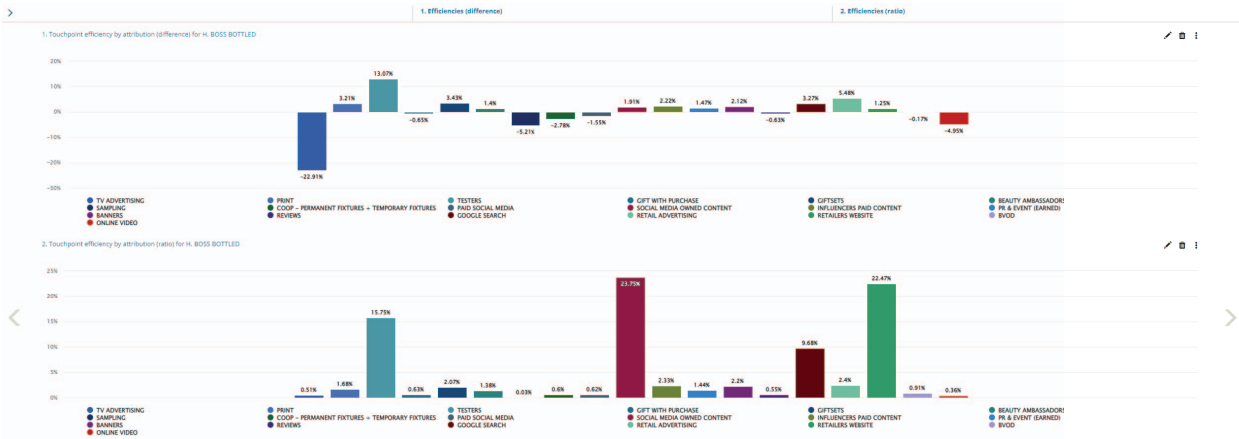

Fig. 2. Screenshot of the Zio Web application, showing media efficiency output charts from what-if scenarios after running agent-based simulations.

brand ($a_i(b) \in \{0, 1\}$); ii) we model the importance of each driver $\omega \in \Omega$ when making their decisions (d_i^ω), and iii) the perception $p_i^\omega(b) \in [0, 1]$ of the agent with respect to each brand $b \in B$ and driver ω is also defined. The latter variables evolve over time during the terminating discrete ABM simulation for a maximum T time-steps (each time-step representing a week). Variables' evolution is modified by a set of touch-points or media channels of the market and the social network [5] connecting the agents of the population among them to facilitate a word-of-mouth process. These touch-points are defined by the budget investment of each brand (e.g., TV ads for given weeks) as well as by internal media dynamics parameters controlling the nature of the media channel.

B. Validation through automatic calibration

The latter media dynamics parameters are an example of parameters that are unknown for stakeholders and modelers. Therefore, they must be tuned or calibrated [6] to reflect the internal dynamics of the market. Zio allows the modeler to define which parameters are calibrated with their restrictions to match the historical data. Specifically, an EA with a local optimizer (i.e. a memetic algorithm) [7] is included to find the most suitable values for those parameters to be calibrated as it is a derivative-free method to evaluate the solution (i.e., the simulation system). Take into account that an auto-calibration process is a validation tool to end up with a ready-to-exploit model but the modeler is the one to validate the market.

C. What-if scenarios and strategy optimization

Once the model is validated, a Zio user can launch what-if scenarios. These scenarios evaluate marketing strategies by simulating them. Examples of these scenarios are modified marketing mixes, the disruption of an incumbent product in the market, or adaptive price policies for a changing market. Figure 2 shows a Zio screenshot with a set of slides having the outputs of different what-if scenarios after being simulated.

Finally, an EA-based strategy optimization module is included to optimize the marketing budget to increase sales or any other key performance indicator as the brand awareness. The procedure can find good solutions (i.e., budget details and media channels to focus) in a reasonable amount of time.

III. Final Remarks

Zio is an application based on AI to recreate competitive markets. Consumers are modeled through ABM simulations, social networks, and marketing science to add realism to a digital twin of the market. Marketers can calibrate the model and launch optimization processes to obtain the most favorable marketing strategies. The application has been tested and used for a diverse set of markets, from fast good consumers to retailing, insurance companies, and automakers, and has shown a solid behavior and robust performance. However, there are still some pitfalls when deploying the digital twin in uncertain markets with vague data. The validation process is still difficult to perform. Other AI methods could ease this task such as checking if the model has enough granularity (i.e., enough virtual agents) for the expected output of the system.

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